

Novel Sulfide-Chloride Solid-State Electrolytes with Tunable Anion Ratio for Highly Stable Solid-State Sodium-Ion Batteries

Zhi Liang Dong, Yi Gan, Vinicius Martins, Xuchun Wang, Bolin Fu, Enzhong Jin, Yingjie Gao, Yang Hu, Xiaoting Lin, Yi Yuan, Colin Turner, Xin Pang, Hamidreza Abdolvand, Yining Huang, Tsun-Kong Sham,* and Yang Zhao*

Solid-state Na batteries (SSNBs) are among the most promising next-generation energy storage devices due to their high energy density, enhanced safety, and cost-efficiency. Achieving high-performance SSNBs depends on the development of solid-state electrolytes (SSEs) with excellent ionic conductivity, wide electrochemical windows, and robust mechanical properties. Sulfide and halide-based Na SSEs have been widely studied in recent years with their respective strengths and limitations. Herein for the first time, a new family of Na-Zr-S-Cl sulfide-chloride Na SSEs with tailored anion compositions is explored. A high ionic conductivity of $4.89 \times 10^{-4} \text{ S cm}^{-1}$ is realized with a Cl-rich structure, attributed to the unique chloride bridging structure and low Na-ion migration barrier. Furthermore, by tuning the ratio between sulfur and chloride anions, two different unique structures are obtained with different surface morphology, thermal stability, mechanical properties, and electrochemical stability. The Cl-deficient structure of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs demonstrates excellent stability for solid-state Na-ion batteries, maintaining a high reversible capacity of over 90 mAh g^{-1} after 600 cycles at 0.1 C. This study offers insights into the design of new Na SSEs, advancing the development of safer and high-performance SSNBs.

vehicles, have attracted increasing attention in recent years because of the availability and low cost of sodium resources.^[1] Current SIBs use liquid electrolytes with organic solvents, which present good ionic conductivity and wettability, however, the flammability and low thermal stability pose significant safety risks.^[2] Solid-state sodium batteries (SSNBs) employing solid-state electrolytes (SSEs) can address the safety issues at their root and realize the long lifetime and high-energy density.^[3]

Solid-state Na electrolytes (SSNEs) acting as both the ion conductor and separator between the cathode and anode, are indispensable components of SSNBs. The ideal SSNEs should possess various features such as high ionic conductivity, wide electrochemical windows, excellent chemical, electrochemical and mechanical stability, and ease of fabrication.^[4] Different types of SSNEs have been developed in the past decades, which can be classified based on the anions in the structure of SSNEs, such as oxide, sulfide, halide, and hydride.^[5]

Among them, sulfide-based SSNEs, such as Na_3PS_4 , Na_3SbS_4 , Na_3BS_3 , $\text{Na}_{11}\text{Sn}_2\text{PS}_{12}$, etc., are considered as promising candidates for potential practical applications because of their good ion conductivity, excellent plasticity, low interfacial resistance and

1. Introduction

Sodium-ion batteries (SIBs), as the alternative choice for large-scale energy storage systems and moderate-range electric

Z. L. Dong, Y. Gan, V. Martins, B. Fu, E. Jin, Y. Gao, Y. Hu, X. Lin, Y. Yuan, C. Turner, H. Abdolvand, Y. Zhao
 Department of Mechanical and Materials Engineering
 University of Western Ontario
 London, ON N6A 5B9, Canada
 E-mail: yzhao628@uwo.ca

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adma.202503107>

© 2025 The Author(s). Advanced Materials published by Wiley-VCH GmbH. This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](#) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

DOI: 10.1002/adma.202503107

X. Wang, Y. Huang, T.-K. Sham
 Department of Chemistry
 University of Western Ontario
 London, ON N6G 2V4, Canada
 E-mail: tsham@uwo.ca

X. Pang
 CanmetMATERIALS
 Natural Resources Canada
 183 Longwood Road South, Hamilton, ON L8P 0A5, Canada

suitable synthesis conditions.^[6] However, one of the biggest challenges for sulfide-based SSNEs is the narrowest electrochemical window among different types of SSNEs.^[7] The oxidation potential for sulfide-based SSNEs is always less than 2.5 V, which is attributed to the relatively low electronegativity of the chalcogen anions.^[8] Because of this, the sulfide-based SSNEs display high chemical and electrochemical reactivity against high voltage and high capacity layered transition metal oxide (TMO) cathodes.^[8] To extend the electrochemical window, especially the oxidation potential, halogen ion doping is an effective approach. In a previous study, a cation (W^{6+}) and anion (Cl^-) co-doping strategy for Na_3SbS_4 SSEs was demonstrated, leading to improved ionic conductivity, wider electrochemical stability windows, and enhanced electrochemical performance in solid-state sodium-sulfur batteries.^[9] It was found that the crystal structure of Na_3SbS_4 SSEs with W/Cl co-doping had almost no change when it was oxidized to 6 V. Sulfide-based SSEs remain highly challenging to pair effectively with high-voltage TMO cathodes. Halide-based SSNEs were revived in the last few years with improved ionic conductivity from 10^{-9} to over 10^{-4} S cm^{-1} .^[10] Na_2ZrCl_6 was the first and most widely studied halide-based SSNE with an ionic conductivity of 10^{-5} S cm^{-1} . Various cation dopants, such as Y, Er, Cr, and Yb have been shown to enhance its ionic conductivity to 10^{-4} S cm^{-1} .^[11] Halogen ion, such as F, was reported to significantly extend the oxidation potential of the Na_2ZrCl_6 SSEs to 5 V.^[12] However, more efforts are necessary to further improve the room-temperature ionic conductivity of halide-based SSNEs and their stability of electrochemical performances with high-voltage TMO cathodes. Recently, a new framework of oxychloride SSEs, such as $LiZrOCl_x$, which incorporate both oxygen and chlorine ions, have been developed for lithium-based SSEs, offering high ionic conductivity, low interfacial resistance and outstanding electrochemical performances.^[13] Unlike traditional anion doping strategies, oxychloride SSEs inherit the advantage of both oxides and halides, benefiting from their high oxygen content. Mixed-anion chemistry is believed to induce a wide range of distortions in Li sites, leading to significantly enhanced ionic conductivity. However, research on mixed-anion SSEs remains in its infancy and is limited to the oxychloride family.^[14] Furthermore, the influence of the anion ratio on the physical, chemical and electrochemical properties of the SSEs remains unclear.

Herein, we report a new family of Na-Zr-S-Cl sulfide-chloride SSNEs with dual anions. By tuning the ratio between sulfur and chloride ions, two distinct phases of $Na_2S-1.3ZrCl_4$ (Cl-deficient) and $Na_2S-3.3ZrCl_4$ (Cl-rich), were obtained, with high ionic conductivities of 3.41×10^{-4} S and 4.89×10^{-4} S cm^{-1} , respectively. The $Na_2S-1.3ZrCl_4$ and $Na_2S-3.3ZrCl_4$ systems display different Zr polyhedral unit cells with different coordination environments. The high ionic conductivity is attributed to the sulfur-chloride bridging structure, a highly disordered Na ion distribution, and a low Na^+ migration barrier. The $Na_2S-1.3ZrCl_4$ SSEs demonstrated a more compact morphology with superior thermal and mechanical properties, compared to $Na_2S-3.3ZrCl_4$ SSEs. Furthermore, $Na_2S-1.3ZrCl_4$ SSEs, with minimum sulfur oxidation, show significantly enhanced chemical and electrochemical interface stability with the Na layer-cathode compared to the $Na_2S-3.3ZrCl_4$ SSEs. As a result, the solid-state Na-ion batteries (SSNIBs) with $Na_2S-1.3ZrCl_4$ SSEs exhibit superior stable performance with a high reversible capacity of over 90 mAh g^{-1}

after 600 cycles at 0.1 C, which is at the top of the reported literature. This research reveals a new family of sulfide-chloride SSEs and opens a new avenue for designing high-performance solid-state Na-ion batteries.

2. Results and Discussion

The $Na_2S-ZrCl_4$ sulfide-chloride SSEs were prepared through the ball-milling method at 550 rpm for 21 h, using Na_2S and $ZrCl_4$ as the starting materials at various stoichiometric ratios ($Na_2S-xZrCl_4$, x represents the molar ratio of $ZrCl_4/Na_2S$ and x ranges from 1 to 3.7). Electrochemical impedance spectroscopy (EIS) results are depicted in Figure S1a (Supporting Information) and the room-temperature ionic conductivities of all SSEs are summarized in Figure 1a. With a 1:1 ratio of Na_2S and $ZrCl_4$ ($x = 1$), the $Na_2S-ZrCl_4$ SSEs presented an ionic conductivity of 1.89×10^{-4} S cm^{-1} , which increased to 3.41×10^{-4} S cm^{-1} when x was increased to 1.3. Further increasing the ratio led to a decrease in conductivity until x reached 2. However, the ionic conductivity improved again, reaching 4.89×10^{-4} S cm^{-1} when x reached 3.3. After that, the ionic conductivity dropped sharply. Different from previous studies of oxychloride, the results suggest that there are potentially two phases formed among the $Na_2S-ZrCl_4$ sulfide-chloride SSEs with optimal ionic conductivities: $Na_2S-1.3ZrCl_4$ SSEs and $Na_2S-3.3ZrCl_4$ SSEs. The ball mill synthesis conditions in terms of ball mill time and speed were further optimized, as shown in Figure S1b–f (Supporting Information). This sulfide-chloride dual anion SSEs demonstrated much higher ionic conductivity than the halide SSEs of Na_2ZrCl_6 (1.8×10^{-5} S cm^{-1}).^[15] For further comparison, the Na-Zr-S sulfide SSEs were synthesized with the same stoichiometric ratios ($Na_2S-1.3ZrS_2$ and $Na_2S-3.3ZrS_2$) and ball mill conditions. Figure S2 (Supporting Information) demonstrates the EIS of the $Na_2S-1.3ZrS_2$ and $Na_2S-3.3ZrS_2$ sulfide SSEs. However, $Na_2S-1.3ZrS_2$ presented large resistance with almost no ionic conduction and the ionic conductivity for $Na_2S-3.3ZrS_2$ was around 10^{-7} S cm^{-1} . Therefore, this novel sulfide-chloride dual anion structure demonstrates a unique Na ion transportation pathway compared to the individual sulfide and halide SSEs. The activation energy determined from Arrhenius plots was 0.387 and 0.351 eV for $Na_2S-1.3ZrCl_4$ and $Na_2S-3.3ZrCl_4$ SSEs, respectively, indicating a low Na^+ migration barrier and allowing rapid Na^+ diffusion (Figure 1b; Figure S3, Supporting Information). In addition, the electronic conductivities of $Na_2S-1.3ZrCl_4$ and $Na_2S-3.3ZrCl_4$ SSEs were found to be lower than 10^{-10} S cm^{-1} , as shown in Figure S4 (Supporting Information). This behaviour suggests that the sulfide-chloride SSEs could serve as an effective electronic insulators for SSNIBs.

The physical properties and thermal stability of $Na_2S-1.3ZrCl_4$ and $Na_2S-3.3ZrCl_4$ SSEs were further investigated. The scanning electron microscope (SEM) was used to examine the surface morphology of $Na_2S-1.3ZrCl_4$ and $Na_2S-3.3ZrCl_4$ SSEs (Figure S5, Supporting Information). The SEM images show continuous and smooth surface morphology for both SSEs, in which the $Na_2S-3.3ZrCl_4$ pellet presents more cracks and grains on the surface. The thermostability of $Na_2S-1.3ZrCl_4$ and $Na_2S-3.3ZrCl_4$ SSEs was studied using thermogravimetric analysis (TGA), as shown in Figure 2c. Notable differences in thermal stability were

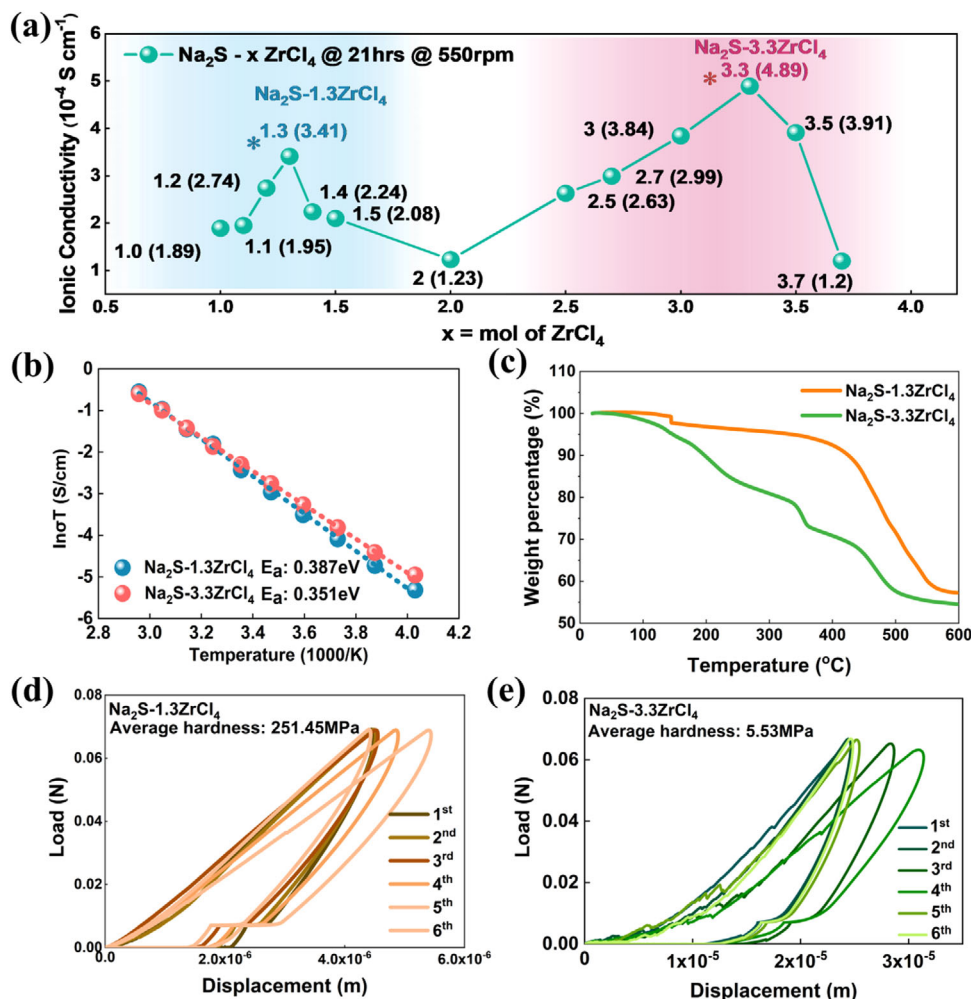


Figure 1. a) The ionic conductivity of Na₂S-*x*ZrCl₄ SSEs (*x* = 1–3.7); b) the Arrhenius plot of Na₂S-1.3ZrCl₄ and Na₂S-3.3ZrCl₄ SSEs; c) The TGA curve of Na₂S-*x*ZrCl₄ SSEs (*x* = 1.3 and 3.3); The nanoindentation load-displacement curve of d) Na₂S-1.3ZrCl₄ SSEs; e) Na₂S-3.3ZrCl₄ SSEs.

observed between the two compositions. The Na₂S-1.3ZrCl₄ SSEs exhibit excellent thermal stability with minimum decomposition until 400 °C. In contrast, the Na₂S-3.3ZrCl₄ SSEs began to decompose at 150 °C, likely due to the poor thermal stability of the Cl-rich structure.^[16] The XRD pattern of Na₂S-3.3ZrCl₄ SSEs at 150 °C suggests the decomposed products of the SSEs were NaCl and there were some other unknown impurities were precipitated as well (Figure S6, Supporting Information).^[17] The mechanical properties of Na₂S-1.3ZrCl₄ and Na₂S-3.3ZrCl₄ SSEs were studied using nanoindentation (Figure 1d,e). It is obvious that the displacement of Na₂S-3.3ZrCl₄ SSEs is around 6 times higher than that of Na₂S-1.3ZrCl₄ SSEs under the load. The hardness of the Na₂S-1.3ZrCl₄ SSEs and Na₂S-3.3ZrCl₄ SSEs, estimated from the average hardness of the two structures, was approximately 251.45 and 5.53 MPa, respectively. This significant difference highlights the superior mechanical properties of the Na₂S-1.3ZrCl₄ SSEs compared to the Cl-rich Na₂S-3.3ZrCl₄ SSEs. The difference in mechanical properties is presumably closely related to the various sulfur and chloride ratios that likely play a critical role in the local structures of the sulfide-chloride SSEs.

The poor mechanical properties observed in other Cl-rich structures further support this correlation.^[18] In summary, both Na₂S-1.3ZrCl₄ and Na₂S-3.3ZrCl₄ SSEs present high ionic conductive (> 10⁻⁴ S cm⁻¹) at room temperature. However, the Na₂S-1.3ZrCl₄ SSEs exhibit a more compact morphology along with better thermal and mechanical properties.

To gain insights into the relationship between local structure and Na⁺ diffusion within the two phases of the sulfide-chloride SSEs, detailed structural analyses were performed. X-ray diffraction (XRD) was used to determine the phases and structure of Na₂S-*x*ZrCl₄ sulfide-chloride SSEs. The XRD patterns of Na₂S-1.3ZrCl₄ and Na₂S-3.3ZrCl₄ SSEs with several references are exhibited in Figure 2a and the XRD pattern for Na₂S-*x*ZrCl₄ with other stoichiometric ratios and ball mill conditions are presented in Figure S7 (Supporting Information). The XRD patterns for all Na₂S-*x*ZrCl₄ sulfide-chloride SSEs demonstrated the main features of a broad peak without obvious lattice peaks, indicating that as-synthesized SSEs are mainly amorphous. However, small peaks at 15° and 31° can be observed for all Na₂S-*x*ZrCl₄ SSEs, suggesting the possible impurity of NaCl. The XRD patterns for Na₂S-1.3ZrCl₄ SSEs and Na₂S-3.3ZrCl₄ SSEs under different

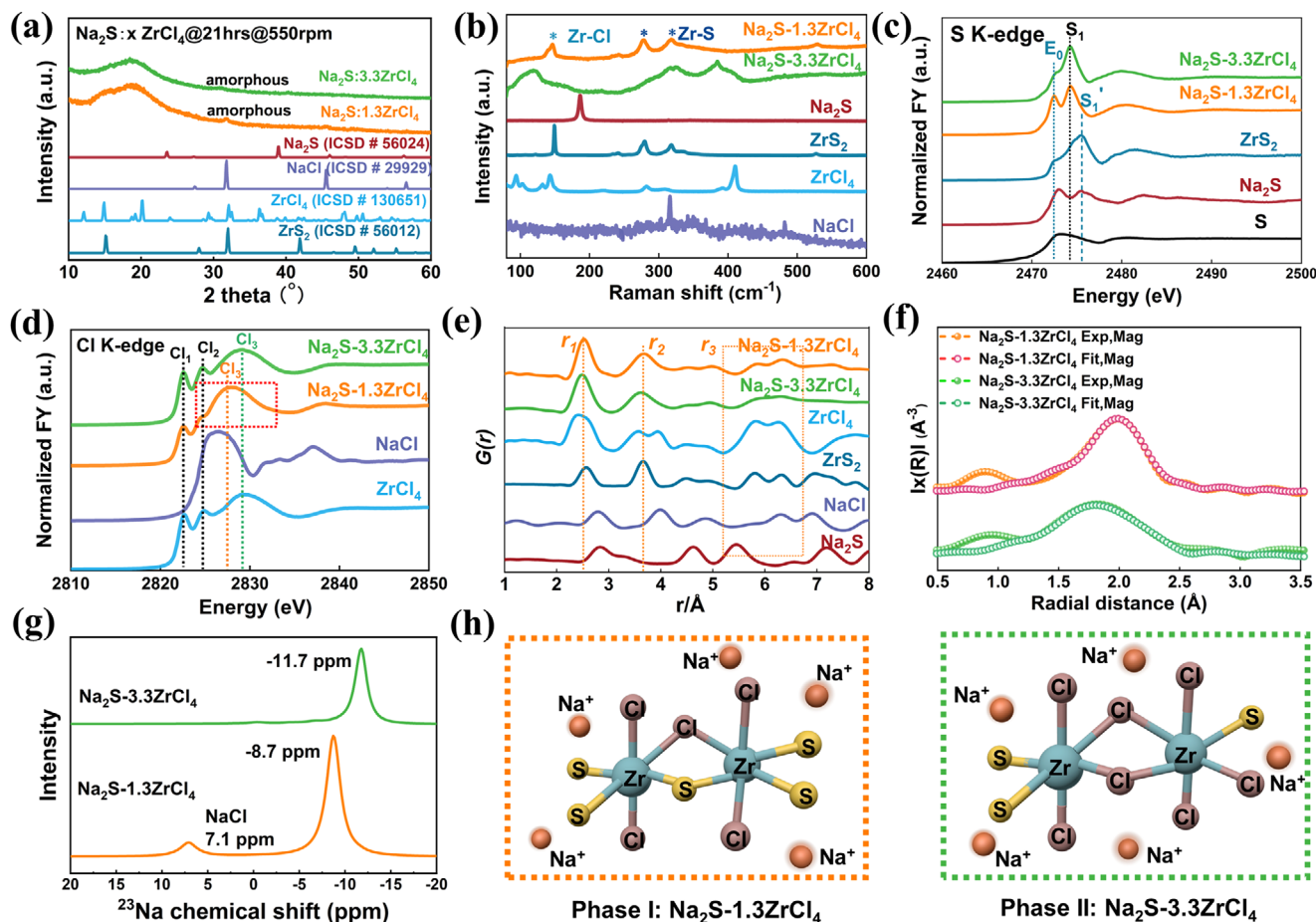


Figure 2. a) The XRD patterns of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3), Na_2S , NaCl , ZrCl_4 , and ZrS_2 ; b) The Raman spectra of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3), Na_2S , NaCl , ZrCl_4 , and ZrS_2 ; c) The S K-edge XANES spectra of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3), Na_2S , ZrS_2 , and S ; d) The Cl K-edge XANES spectra of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3), NaCl and ZrCl_4 ; e) The PDF spectra of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3), Na_2S , NaCl , ZrCl_4 and ZrS_2 ; f) FT-EXAFS fitting results for the Zr K-edge spectrum of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3); g) ^{23}Na MAS NMR spectra of the $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. Spectra were acquired at 600 MHz with a spinning rate of 22 kHz; h) The scheme diagram of two Zr-based polyhedron unit structures: $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs.

ball mill conditions are presented in Figure S7b–d (Supporting Information). Raman spectroscopy was further used to understand the structures. Figure 2b shows the Raman spectra for both $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. For $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs, there are three significant peaks located at 146.03, 278.33, and 317.14 cm^{-1} . The strong peak at 146.03 cm^{-1} belongs to the octahedral $[\text{ZrCl}_6]^{2-}$ unit from the original ZrCl_4 structure.^[19] The peaks at 278.33 and 317.14 cm^{-1} are attributed to the E_g and A_{1g} peaks, respectively, suggesting the out-of-plane and in-plane sketching from the $[\text{Zr-S}_2-\text{Zr-S}_2]$ structure.^[20] In addition, the Raman spectra exhibit a prominent A_{1g} peak feature and the absence of E_g peaks, indicating out-of-plane stretching from the $[\text{Zr-S}_2-\text{Zr-S}_2]$ structure and the lack of in-plane stretching.^[20]

To study the local chemical structure of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ sulfide-chloride SSEs, X-ray absorption near edge structure (XANES) was conducted. Figure 2c shows the S K-edge XANES spectra for both $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. Both SSEs have the same E_0 position as the ZrS_2 , indicating a similar oxidation state of S in both SSEs. However, the S K-edge characteristics of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs are different; in that

both Na_2S and ZrS_2 features are observed for $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs and only ZrS_2 feature is present for $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. The ZrS_2 feature demonstrates the formation of $[\text{Zr-S}_2-\text{Zr-S}_2]$ structures bridging by sulfur atoms.^[20b] In addition, the S_1 peak position for both SSEs also demonstrates a lower energy shift compared with the S_1' peaks of Na_2S and ZrS_2 references. The $\text{S } 1s \rightarrow p$ and d hybridization photoelectron transitions in both SSEs are affected by Zr^{4+} than Na^+ . The amount of Zr-Cl can influence the local chemical environment of Zr-S bonding in Zr^{4+} polyhedral and $\text{S } 1s \rightarrow 2p, 3d$ orbital excitations.^[17,21] Figure 2d displays the Cl K-edge XANES spectra of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. By comparing the Cl_1 peak position and the line shapes, it can be seen that the primary neighbouring bonding element of Cl is Zr for both structures. For $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs, a shoulder instead of a peak was observed at Cl_1 position with a lower energy shift compared to ZrCl_4 . This indicates that the increased presence Zr-S bonds in $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs changes the local chemical environment of Zr-Cl bonds, resulting in longer Zr-Cl bond lengths compared to the Cl-rich phase of $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$.

To investigate the local chemical environment of the central atom of Zr in $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs, Pair Distribution Function (PDF), Zr K-edge and $\text{L}_{3,2}$ -edge XANES are carried out (Figure 2e; Figures S8 and S9, Supporting Information). Figure 2e demonstrates the PDF results of both $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs, showing the presence of Zr–S and Zr–Cl bonds in both compositions. Additionally, the Zr–Zr peaks observed around 3.5 and 3.6 Å mainly correspond to the S and Cl-edge connection of Zr-centered polyhedron.^[17,22] Figure S8a (Supporting Information) presents the Zr K-edge XANES spectra of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. Based on both Zr_1 and Zr_1' peak positions of the two SSEs, it is observed that the unit cell of Zr polyhedron in $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ is related to the ZrS_2 and that of $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ is closer to $[\text{ZrCl}_6]^{2-}$, which is consistent with the above S K-edge and Cl K-edges results. The Zr $\text{L}_{3,2}$ -edge XANES spectra of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs are shown in Figure S9a (Supporting Information). The feature of $\text{L}_{3,2}$ -edge is correlated to the $2p \rightarrow e_g$ and $2p \rightarrow t_{2g}$ transitions, respectively.^[23] The magnitude of the multiple effects in $\text{L}_{3,2}$ -edge is relevant to the coordination number of the Zr polyhedron,^[24] indicating that the coordinate number of the Zr polyhedral unit cell is the same for both structures. To further quantitatively understand the coordination environment around the central Zr in $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs, Zr K-edge FT-EXAFS spectroscopy was subsequently utilized as shown in Figure 2f and Figure S8d (Supporting Information). The EXAFS fitting provides the average coordination information nearest to Zr in both SSEs. According to the fitting results, a central Zr atom is surrounded by 3.0 sulfur atoms at ≈ 2.65 Å and 4.1 chloride atoms at ≈ 2.5 Å for $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs (Table S1, Supporting Information). In addition, the central Zr atom is surrounded by 2.6 sulfur atoms at ≈ 2.61 Å and 6.2 chloride atoms at ≈ 2.41 Å in $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs (Table S2, Supporting Information).

To clarify the local chemical and electronic environment surrounding the sodium ions, ^{23}Na solid-state nuclear magnetic resonance (NMR) spectroscopy was utilized. The ^{23}Na magic angle spinning (MAS) spectra of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs are shown in Figure 2g, with the corresponding NMR spectral parameters listed in Table S3 (Supporting Information). In the spectrum of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$, two resonances are observed at 7.1 and -8.7 ppm. These central transitions do not display a typical quadrupolar lineshape, with well-defined spectral parameters such as quadrupolar coupling constant (C_Q) and asymmetry parameter (η_Q). Instead, they present a relatively narrow Gaussian lineshape, with full width at half maximum (FWHM) of 350 and 280 Hz, respectively. This suggests that the sample has a local structural disorder with variations in the electric field gradient (EFG) that causes a spread of quadrupolar coupling constants, therefore leading to a broadening that resembles a Gaussian profile. This is consistent with the amorphous nature of the material. The signal at 7.1 ppm, with an intensity of 2%, can be assigned to trace the impurity of NaCl.^[25] This is confirmed by the X-ray diffraction results. The second signal with a lower chemical shift of -8.7 ppm is attributed to the sodium ions in the $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ sample. This chemical shift is typical of compounds in which sodium ions adopt a high coordination number, generally between five and six.^[26] Notably, the absence of the Na_2S signal at ≈ 49 ppm^[27] suggests that all Na_2S was consumed during syn-

thesis. Furthermore, it suggests that the sulfur atoms are mostly coordinated to Zr rather than Na, which is consistent with the findings from the S K-edge XANES results. In the spectrum of $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$, a single resonance appears at -11.7 ppm with a slightly smaller FWHM of 240 Hz. The decrease in the chemical shift from -8.7 to -11.7 ppm suggests an increased coordination number around the Na^+ cations, accompanied by shorter Na–ligand bonds.^[26] The decrease in chemical shift has been found to be correlated with the shortening of the bonds between Na^+ ions and the ligands.^[28]

As aforementioned, the Zr polyhedron is the main unit cell structure for both $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. It consists of $[\text{ZrCl}_6]^{2-}$ and $[\text{ZrS}_x\text{Cl}_y]$ units, which are connected via corner-shared sulfur and chlorine atoms. Highly disordered Na^+ ions with high mobilities are distributed around the Zr polyhedron structures. In this sulfide-chloride SSEs, sulfur is predominantly coordinated with Zr^{4+} instead of Na^+ , existing as bridging and non-bridging sulfur atoms. This unique configuration provides a relatively open structure with free volume, enabling efficient Na mobility. It also can create a distorted sodium-site environment and provide percolating pathways for sodium transportation with a low energy barrier.^[17,29] Furthermore, the presence of sulfur as a weak trap for Na^+ ions facilitates their release from Coulomb traps, enabling effective Na^+ mobility between its initial and final sites.^[17,30] In addition, $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs display different sulfur and chloride ratios, forming Cl-lean and Cl-rich structures that lead to different thermal and mechanical properties (Figure 2h). The Cl-deficient structure ($\text{Na}_2\text{S}-1.3\text{ZrCl}_4$) provides more sulfur bridging bonds, potentially enabling better stability compared to the chlorine bridging bonds in the Cl-rich structure of $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$.

The Zr K-edge, Zr $\text{L}_{3,2}$ -edge, Cl K-edge and S K-edge XANES spectra for other $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs under different synthesis conditions are presented in Figures S8–S11 (Supporting Information). There is no obvious difference among all samples in the Zr K-edge, Zr $\text{L}_{3,2}$ -edge and Cl K-edge XANES spectra (Figures S8–S10, Supporting Information), indicating the formation of the same Zr polyhedron structures. In addition, the white-line shape of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs in Figure S11a (Supporting Information) exhibits features similar to those of either $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ or $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs, suggesting that the local chemical environment of sulfur in Na–Zr–S–Cl SSEs closely resembles these specific compositions. However, there are some significant differences were observed in S K-edge spectra for the samples from different synthesis conditions (Figure S11b–d, Supporting Information). As shown in Figure S11b (Supporting Information), the white-line shapes of the two SSEs under varying ball-milling speeds differs from the feature observed in Figure 2c, indicating that ball-milling speed significantly influences the local chemical environment of sulfur within Zr-centered polyhedral. Comparing the white-line shapes of the SSEs under different reaction times (Figure S11c,d, Supporting Information), $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs still exhibit a white-line shape similar to ZrS_2 , suggesting that the $[\text{Zr}-\text{S}_2-\text{Zr}-\text{S}_2]$ structures remain intact. In contrast, $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs show a distinct white-line shape compared to ZrS_2 , particularly after 27 h of reaction, indicating the absence of sixfold ZrS_2 structures. In summary, the S K-edge XANES spectra of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ sulfide-chloride SSEs reveal that the local sixfold ZrS_2 structures within Zr-centered polyhedral

are highly sensitive to synthesis conditions, especially in Cl-rich compositions.

To understand the influence of the two different unit structures on the electrochemical properties of $\text{Na}_2\text{S-ZrCl}_4$ sulfide-chloride SSEs, various electrochemical characterizations were performed, including linear sweep voltammetry (LSV), cyclic voltammetry (CV), symmetric cell testing and SSNIBs evaluations. In addition, the schematic diagram of the cell configuration for each electrochemical characterization is displayed in Figure S12 (Supporting Information). The LSV was used to investigate the electrochemical window of $\text{Na}_2\text{S-1.3ZrCl}_4$ and $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs, as shown in Figure S13 (Supporting Information). In the LSV cell, carbon black (CB) and SSE (mass ratio: 3: 7) composites were used as the working electrode, while a Na-Sn alloy served as the counter/reference electrode. A thin Na_3PS_4 layer was used to prevent direct contact and side reaction between the $\text{Na}_2\text{S-ZrCl}_4$ SSEs and Na-Sn anode. For $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs, the potential scanned from the open-circuit voltage (OCV) to 5 V, with the SSEs beginning to decompose at ≈ 3.43 V. This decomposition potential is significantly higher than the oxidation potential of ≈ 2.5 V observed in traditional Na sulfide SSEs, such as Na_3PS_4 and Na_3SbS_4 . The $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs presented a similar oxidation potential of 3.63 V. This suggests that the Cl^- anion could suppress the oxidation of Zr-S bonds at high potentials, resulting in remarkably improved electrochemical stability and potential compatibility with high-voltage cathodes. When the potential reduced from OCV to 1.0 V, both $\text{Na}_2\text{S-1.3ZrCl}_4$ and $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs showed obvious decomposition at ≈ 2.5 V, indicating relatively poor stability on the anode. The inferior anode stability is a common issue for most sulfide and halide-based Na SSEs.^[9,14,17,21] To further study the anode stability of both SSEs, the symmetric cells of Na/SSEs/Na were tested. Figure S14 (Supporting Information) exhibits the symmetric battery performance of $\text{Na}_2\text{S-1.3ZrCl}_4$ and $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs obtained under different areal capacities of 0.1 and 0.5 mAh cm^{-2} while maintaining the same current density of 0.5 mA cm^{-2} . In the initial cycles, the symmetric cell with $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs showed three times higher overpotential compared to the $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs-based cell. It suggests that the $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs would react with the Na metal foil immediately forming the interface with large resistances. The continuous reactions between the $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs and Na foil result in a thick interface layer with increased overpotential. In contrast, the symmetric cell with $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs exhibited a slightly increased overpotential in the initial cycles, indicating a more stable interface formed between the $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs and Na metal anode. In addition, the $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs demonstrate a long lifetime with low overpotential in symmetric cells compared to the $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs.

To showcase the application of the new sulfide-chloride SSEs, room-temperature SSNIBs were manufactured and tested. As shown in Figure S12 (Supporting Information), either the $\text{Na}_2\text{S-1.3ZrCl}_4$ or $\text{Na}_2\text{S-3.3ZrCl}_4$ was used as the main SSEs layer. The cathode composites consist of $\text{Na}_{0.85}\text{Mn}_{0.5}\text{Ni}_{0.4}\text{Fe}_{0.1}\text{O}_2$ (NMNFO), sulfide-chloride SSEs (either $\text{Na}_2\text{S-1.3ZrCl}_4$ or $\text{Na}_2\text{S-3.3ZrCl}_4$) and conductive carbon. A Na_3PS_4 interlayer was used between the SSEs and Na-Sn alloy to prevent undesirable reactions. Figure S15 (Supporting Information) demonstrates the CV profiles for the SSNIB using $\text{Na}_2\text{S-1.3ZrCl}_4$ and $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs at a scan rate of 0.2 mV S^{-1} for 5 cycles. The SSNIBs with $\text{Na}_2\text{S-}$

1.3ZrCl_4 SSEs exhibit similar redox reaction peaks compared to previous works. However, the SSNIBs with $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs present an almost undetectable response current compared to the $\text{Na}_2\text{S-1.3ZrCl}_4$ -SSNIBs, indicating the poor compatibility and possible side reactions between the NMNFO cathode and $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs during electrochemical cycling. Nevertheless, the SSNIBs with $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs also demonstrate good electrochemical reversibility with overlaps of the oxidation and reduction peaks in CV.

The electrochemical performances of the SSNIBs using the two sulfide-chloride SSEs are demonstrated in Figure 3a–d, Figures S16–S18 (Supporting Information). The diameter of cell molds is approximately 1 cm, and the area of the mold's surface is approximately 0.785 cm^2 . Figure 3a exhibits the charge-discharge profiles for the $\text{Na}_2\text{S-1.3ZrCl}_4$ -SSNIBs at 0.1 C (1 C = 120 mAh g^{-1}). The initial reversible discharging capacity is 120 mAh g^{-1} , and the Coulombic efficiency is 100%, indicating excellent compatibility and stability between the $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs and NMNFO cathode. In contrast, the $\text{Na}_2\text{S-3.3ZrCl}_4$ -SSNIBs present an extremely low discharging capacity (2 mAh g^{-1}), indicating strong side reactions between the $\text{Na}_2\text{S-3.3ZrCl}_4$ SSEs and NMNFO cathode (Figure 3b). The charge-discharge performance is consistent with the CV results. As discussed above, the local chemical environment of the two different unit structures has strong effects on the electrochemical reactions between the SSEs and NMNFO cathode during the oxidation-reduction process. The $\text{Na}_2\text{S-1.3ZrCl}_4$ SSNIBs demonstrate excellent electrochemical cycling stability at 0.1 C, as illustrated in Figure 3c. The $\text{Na}_2\text{S-1.3ZrCl}_4$ -SSNIBs can maintain a high reversible capacity of 92.81 mAh g^{-1} after 600 cycles with a Coulombic efficiency of 99.26%, and the capacity retention is around 76.6%. Furthermore, the cycling performances of the $\text{Na}_2\text{S-1.3ZrCl}_4$ -SSNIBs were tested at a higher current density (0.3 C) (Figure 3d; Figure S16a, Supporting Information). Figure S16a (Supporting Information) displays the charge-discharge profiles for $\text{Na}_2\text{S-1.3ZrCl}_4$ SSNIBs at 0.3 C. The first three cycles were tested under 0.1 C for activation, and the following cycles were measured under 0.3 C. The initial reversible discharging capacity measured at 0.3 C was 83.96 mAh g^{-1} , with Coulombic efficiency of $\approx 90\%$. Remarkably, the $\text{Na}_2\text{S-1.3ZrCl}_4$ -SSNIBs demonstrate superior long-cycling stability and slow capacity fading, with specific capacity of 52.44 mAh g^{-1} after 1,400 cycles. Figure S16b,c (Supporting Information) display the rate performance of $\text{Na}_2\text{S-1.3ZrCl}_4$ -SSNIBs, where reversible discharge capacities of 105.41, 88.91, 69.39, and 53.17 mAh g^{-1} were achieved at 0.1 C, 0.2 C, 0.3 C, and 0.4 C, respectively. For comparison, the $\text{Na}_2\text{S-1.3ZrCl}_4$ -based SSNIBs without NPS interlayer were also tested under 0.1 C. The 1st specific charging capacity is 119.36 mAh g^{-1} , and it almost cannot be discharged (Figure S17a, Supporting Information), indicating the severe side reaction that happened between $\text{Na}_2\text{S-1.3ZrCl}_4$ SSEs and Na-Sn anode. In addition, the EIS results also show a large resistance from the insulating interfaces (Figure S17b, Supporting Information). In contrast, the EIS results of $\text{Na}_2\text{S-1.3ZrCl}_4$ -based SSNIBs with an NPS interlayer suggest a much smaller resistance, indicating the stable interfaces formed with the NPS interlayer (Figure S17c, Supporting Information). To comprehend the impact of varying masses of composite cathodes on their performance, the rate performance of $\text{Na}_2\text{S-1.3ZrCl}_4$ SSNIBs with a 12 mg composite cathode was evaluated at 0.1 C to 0.5 C (where 1

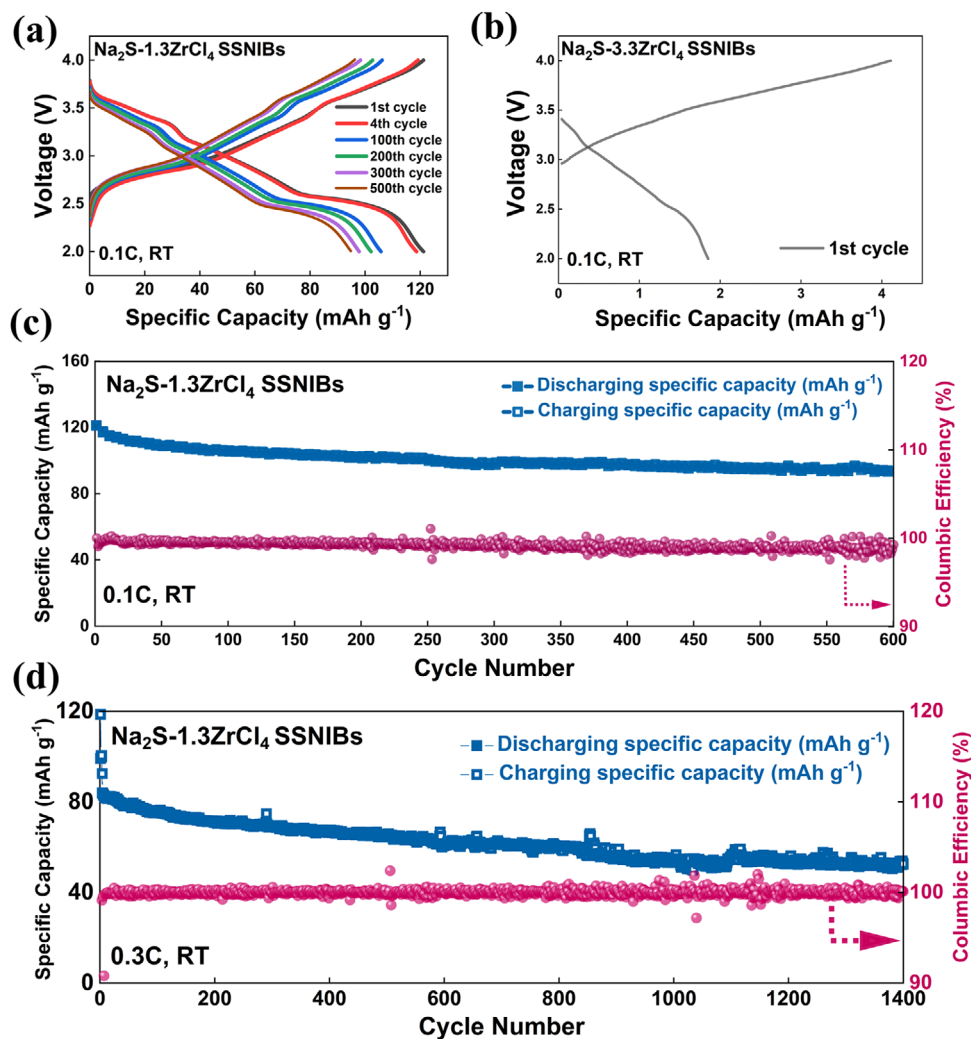


Figure 3. a) The charge-discharge profiles for $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSNIBs at 0.1 C; b) The 1st cycle of charging and discharging pattern of $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSNIBs at 0.1 C; The cycling performances and Coulombic efficiency for $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSNIBs at c) 0.1 C and d) 0.3 C at room temperature.

C equals 120 mAh g^{-1} (Figure S18, Supporting Information). Figure S18b (Supporting Information) illustrates the rate performance of these SSNIBs, revealing reversible discharge capacities of 138.32, 123.23, and $101.24 \text{ mAh g}^{-1}$ at 0.1 C, 0.2 C, and 0.3 C, respectively. The low ionic conductivity of the NPS interlayer significantly limited the achievable capacities at higher rates. Nevertheless, the cell maintained 93.2% ($128.92 \text{ mAh g}^{-1}$) of its initial capacity when the C rate was reverted to 0.1 C, further confirming the excellent electrochemical stability and compatibility between $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs and NMNFO cathodes in SSNIBs.

For the halide-based SSEs, such as Na-Zr-Cl, and Na-Zr-Y-Cl, NaCrO_2 has been widely used as the cathode material. However, it presents lower capacity, low potential and poor cycling stability compared to NMNFO.^[11c,31] Furthermore, compared to the recently developed oxychlorides SSEs (Na-Ta-O-Cl, Na-Hf-O-Cl, etc.),^[14,17] sulfide-chloride SSEs used in our work demonstrated even better cycling stability and capacity retention. Compared to the electrochemical performances reported in the literature,^[11c,14,17,31] the $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ sulfide-chloride SSEs de-

veloped in this work demonstrate higher capacity and excellent cycling stability, outperforming the previously reported results.

To understand the difference in the electrochemical behavior between $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs in SSNIBs, the interfacial chemistry on the cathode composite was analyzed using XANES spectra before and after the first charge-discharge cycle. Figure 4b presents the S K-edge XANES spectra for $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3), $\text{Na}_2\text{S}-x\text{ZrCl}_4$ SSEs ($x = 1.3$ and 3.3) cathode composite before cycling and after the 1st charge and discharge. As shown in Figure 4b, SO_4^{2-} peaks at the energy of $\approx 2483 \text{ eV}$ were observed for both SSEs when they were mixed with the cathode before cycling, indicating a chemical reaction between the SSEs and the NMNFO cathode. This reaction results in sulfur oxidation within the SSEs. However, for $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs, the SO_4^{2-} peak disappeared after the first charging process, suggesting the reduction of SO_4^{2-} during the electrochemical reactions. In contrast, the intensity of the SO_4^{2-} peaks in the $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs increased after the first charging cycle, indicating an enhanced side reaction between the $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs and the NMNFO cathode during the electrochemical process. In

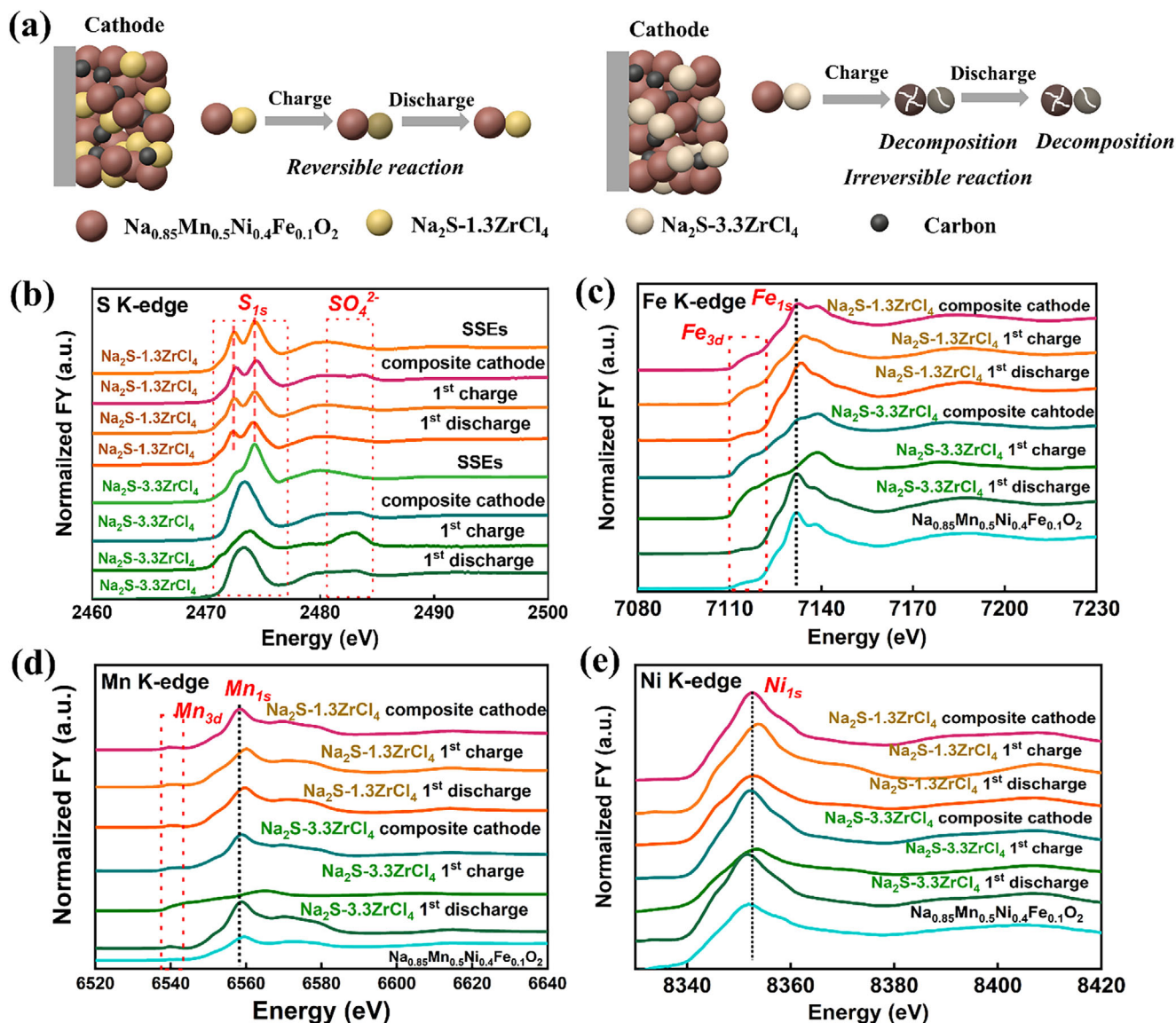


Figure 4. a) The schematic diagram of Na₂S-*x*ZrCl₄ SSEs (*x* = 1.3 and 3.3) cathode composite after the 1st cycle; b) The S K-edge XANES spectra for Na₂S-*x*ZrCl₄ SSEs (*x* = 1.3 and 3.3) cathode composite before cycling and after the 1st charge and discharge; c) The Fe K-edge XANES spectra for Na₂S-*x*ZrCl₄ SSEs (*x* = 1.3 and 3.3) cathode composite before cycling and after the 1st charge and discharge; d) The Mn K-edge XANES spectra for Na₂S-*x*ZrCl₄ SSEs (*x* = 1.3 and 3.3) cathode composite before cycling and after the 1st charge and discharge; e) The Ni K-edge XANES spectra for Na₂S-*x*ZrCl₄ SSEs (*x* = 1.3 and 3.3) cathode composite before cycling and after the 1st charge and discharge.

addition, differences in the white line at the S_{1s} region between the Na₂S-1.3ZrCl₄ and Na₂S-3.3ZrCl₄ SSEs further suggest distinct interfacial stability between the two different unit structures. For Na₂S-1.3ZrCl₄ SSEs, the white line feature of S_{1s} orbital is identical to the Na₂S-1.3ZrCl₄ composite cathode, suggesting the local chemical environment of S remains stable and general features of Na₂S-1.3ZrCl₄ SSEs unit cell remained. However, for Na₂S-3.3ZrCl₄ SSEs, the white line feature of S_{1s} orbital is obviously different from the Na₂S-3.3ZrCl₄ composite cathode. The ZrS₂ unit structure disappears, and a single peak of S can be observed due to the oxidation of sulfur in the Zr polyhedron, indicating not only SO₄²⁻ bonding formed, but also the local unit cell was broken.

Figure 4c–e demonstrates the Fe, Mn, and Ni K-edges XANES spectra for the Na₂S-1.3ZrCl₄ and Na₂S-3.3ZrCl₄ SSEs in cathode composites before and after the first cycle. The pristine NMNFO was used as a reference in each spectrum. In Figure 4c, the Fe_{1s} peak position of the Fe K-edge XANES spectra remains unchanged, indicating that the NMNFO cathode is relatively stable with the Na₂S-1.3ZrCl₄ SSEs. Additionally, a shoulder is observed in the pre-edge of the Fe K-edge spectra for the cathode composition with Na₂S-1.3ZrCl₄ SSEs, indicating Fe 3d orbital occupation. This feature diminishes after the first cycle, indicating free electron movement during the electrochemical reactions. Similar behavior is observed for the Mn and Ni K-edges (Figure 4d,e). In contrast, Fe, Mn, and Ni K-edges of the cathode composite

with $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs show a significantly different white line compared to that of the NMNFO reference, indicating that both chemical and electrochemical reactions occurred on the NMNFO cathode in the $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs-cathode composites. Figure S19 (Supporting Information) displays the Zr L_3 -edge and Cl K-edge XANES spectra of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs in the cathode composite before and after the first cycle. There are some peak ratio changes between Zr_1 and Zr_2 of Zr L_3 -edge XANES in both SSEs composite cathodes after the 1st discharge (Figure S19a, Supporting Information). In addition, the peak ratio between Cl_1 and Cl_2 of Cl K-edge XANES in $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs-cathode has changed after 1st (Figure S19b, Supporting Information). These phenomena suggest there might be a hybridized change in the electronic state of d-orbitals of Zr in both SSEs composite and some hybridized changes in the electronic state of Cl in $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs cathode composite during the charging and discharging. As illustrated in Figure 4a, due to the sulfur oxidation in $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs and the structure change in NMNFO, the SSNIBs with $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs present low reversible capacity. However, $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs demonstrate a stable interface with the NMNFO cathode, resulting in high capacity and excellent stability of the SSNIBs.

To understand the structural changes of NMNFO in SSNIBs, the XRD measurement of $\text{Na}_2\text{S}-x\text{ZrCl}_4$ ($x = 1.3$ and 3.3) composite cathodes after mixing and 1st charge-discharge cycle was carried out (Figure S20, Supporting Information). From the XRD results, the lattice structures of NMNFO in $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ composite cathode are relatively stable and there are almost no changes after mixing and 1st charge-discharge cycle. However, the structures of NMNFO in $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ composite cathode are totally affected and changed after mixing and 1st charge-discharge cycle. The XRD results are consistent with the XANES results discussed above, indicating the stable interfaces between the $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs and NMNFO cathodes.

3. Conclusion

In summary, this work successfully demonstrates a new family of Na-Zr-S-Cl sulfide-chloride SSEs with tunable ratios of sulfur and chlorine anions. Two different unit structures were obtained with optimal ionic conductivities ($>10^{-4} \text{ S cm}^{-1}$), higher than that of the individual halide and sulfide SSEs. The two phases of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ and $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ demonstrate similar sulfur and chloride bridging structures but differ in the coordination environments of the Zr polyhedral. The unique structures allow a low Na^+ migration barrier, resulting in high ionic conductivity. The two different structures lead to different surface morphology, thermal stability, mechanical properties and electrochemical stability. The Cl-deficient structure of $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs demonstrates better thermal and mechanical properties and significantly enhanced chemical and electrochemical interface stability with a Na layer-cathode compared to the Cl-rich structure of $\text{Na}_2\text{S}-3.3\text{ZrCl}_4$ SSEs. When using the $\text{Na}_2\text{S}-1.3\text{ZrCl}_4$ SSEs for SSNIBs, a high reversible capacity of 92.81 mAh g^{-1} can be maintained after 600 cycles at 0.1 C and an extremely long cycling life of 1400 cycles can be achieved at 0.3 C. The electrochemical performance achieved in this work using the Na layer-cathode surpasses that reported in the literature. This study not only in-

troduces a new family of sulfide-chloride SSEs but also offers valuable insights and design principles for mixed-anion systems, paving the way for future electrolyte development.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

Z.L.D. and Y.G. contributed equally to this work. The authors thank the support from the Natural Science of Engineering Research Council of Canada (NSERC), the Canada Foundation of Innovation (CFI), and the University of Western Ontario. The synchrotron research was conducted at the SXRMB, HXMA, and BXDS-WHE beamlines of the Canadian Light Source (CLS), which was supported by the Canada Foundation for Innovation (CFI), the Natural Sciences and Engineering Research Council (NSERC), the National Research Council (NRC), the Canadian Institutes of Health Research (CIHR), the Government of Saskatchewan, the University of Saskatchewan. The authors also gratefully acknowledge the great help from Dr. Mohsen Shakouri and Dr. Qunfeng Xiao at the SXRMB beamline, Dr. Ning Chen at the HXMA beamline, and Dr. Graham King at the BXDS-WHE beamline, at the Canadian Light Source.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

Dual-Anion Framework, solid-state electrolytes, Solid-state Sodium-ion Batteries, Sulfide-chloride

Received: February 14, 2025
Revised: April 8, 2025
Published online: May 19, 2025

- [1] a) A. Rudola, R. Sayers, C. J. Wright, J. Barker, *Nat. Energy* **2023**, *8*, 215; b) Y. Wu, W. Shuang, Y. Wang, F. Chen, S. Tang, X.-L. Wu, Z. Bai, L. Yang, J. Zhang, *Electrochem. Energy Rev.* **2024**, *7*, 17; c) W. Zuo, A. Innocenti, M. Zarrabeitia, D. Bresser, Y. Yang, S. Passerini, *Acc. Chem. Res.* **2023**, *56*, 284.
- [2] a) Y. Dong, P. Wen, H. Shi, Y. Yu, Z. S. Wu, *Adv. Funct. Mater.* **2023**, *34*, 2213584; b) S. Zhao, H. Che, S. Chen, H. Tao, J. Liao, X.-Z. Liao, Z.-F. Ma, *Electrochem. Energy Rev.* **2024**, *7*, 3.
- [3] a) Z. Huang, P. Jaumaux, B. Sun, X. Guo, D. Zhou, D. Shanmukaraj, M. Armand, T. Rojo, G. Wang, *Electrochem. Energy Rev.* **2023**, *6*, 21; b) Y. Lu, L. Li, Q. Zhang, Z. Niu, J. Chen, *Joule* **2018**, *2*, 1747.
- [4] a) W. Hou, X. Guo, X. Shen, K. Amine, H. Yu, J. Lu, *Nano Energy* **2018**, *52*, 279; b) Y. Yang, S. Yang, X. Xue, X. Zhang, Q. Li, Y. Yao, X. Rui, H. Pan, Y. Yu, *Adv. Mater.* **2024**, *36*, 2308332.

- [5] C. Zhao, L. Liu, X. Qi, Y. Lu, F. Wu, J. Zhao, Y. Yu, Y. S. Hu, L. Chen, *Adv. Energy Mater.* **2018**, 8, 1703012.
- [6] a) X. Feng, H. Fang, P. Liu, N. Wu, E. C. Self, L. Yin, P. Wang, X. Li, P. Jena, J. Nanda, D. Mitlin, *Angew. Chem., Int. Ed.* **2021**, 60, 26158; b) F. Tsuji, A. Nasu, C. Hotehama, A. Sakuda, M. Tatsumisago, A. Hayashi, *Mater. Adv.* **2021**, 2, 1676; c) H. Wang, Y. Chen, Z. D. Hood, G. Sahu, A. S. Pandian, J. K. Keum, K. An, C. Liang, *Angew. Chem., Int. Ed.* **2016**, 55, 8551; d) Z. Yu, S.-L. Shang, Y. Gao, D. Wang, X. Li, Z.-K. Liu, D. Wang, *Nano Energy* **2018**, 47, 325.
- [7] G. Deysher, Y. T. Chen, B. Sayahpour, S. W. Lin, S. Y. Ham, P. Ridley, A. Cronk, E. A. Wu, D. H. S. Tan, J. M. Doux, J. A. S. Oh, J. Jang, L. H. B. Nguyen, Y. S. Meng, *ACS Appl. Mater. Interfaces* **2022**, 14, 47706.
- [8] V. Lacivita, Y. Wang, S.-H. Bo, G. Ceder, *J. Mater. Chem. A* **2019**, 7, 8144.
- [9] Z. L. Dong, Y. Yuan, V. Martins, E. Jin, Y. Gan, X. Lin, Y. Gao, X. Hao, Y. Guan, J. Fu, X. Pang, Y. Huang, Q. H. Tu, T.-K. Sham, Y. Zhao, *Nano Energy* **2024**, 128, 109871.
- [10] a) Z. Wei, L. F. Nazar, J. Jabnek, *Batteries Supercaps* **2024**, 7, 202400005; b) K. Motohashi, H. Tsukasaki, A. Sakuda, S. Mori, A. Hayashi, *ACS Mater. Lett.* **2024**, 6, 1178; c) Z. Huang, S. Yoshida, H. Akamatsu, K. Hayashi, S. Ohno, *ACS Mater. Lett.* **2024**, 6, 1732; d) J. Park, J. P. Son, W. Ko, J.-S. Kim, Y. Choi, H. Kim, H. Kwak, D.-H. Seo, J. Kim, Y. S. Jung, *ACS Energy Lett.* **2022**, 7, 3293.
- [11] a) L. Wang, Z. Song, X. Lou, Y. Chen, T. Wang, Z. Wang, H. Chen, W. Yin, M. Avdeev, W. H. Kan, B. Hu, W. Luo, *Small* **2024**, 20, 2400195; b) E. Sebt, J. Qi, P. M. Richardson, P. Ridley, E. A. Wu, S. Banerjee, R. Giovine, A. Cronk, S.-Y. Ham, Y. S. Meng, S. P. Ong, R. J. Clément, *J. Mater. Chem. A* **2022**, 10, 21565; c) E. A. Wu, S. Banerjee, H. Tang, P. M. Richardson, J. M. Doux, J. Qi, Z. Zhu, A. Grenier, Y. Li, E. Zhao, G. Deysher, E. Sebt, H. Nguyen, R. Stephens, G. Verbist, K. W. Chapman, R. J. Clement, A. Banerjee, Y. S. Meng, S. P. Ong, *Nat. Commun.* **2021**, 12, 1256; d) R. Schlem, A. Banik, M. Eckardt, M. Zobel, W. G. Zeier, *ACS Appl. Energy Mater.* **2020**, 3, 10164.
- [12] J. Park, D. Han, J. P. Son, H. Kwak, W. Ko, C. Park, C. Lee, H.-W. Lee, J. Kim, K.-W. Nam, Y. S. Jung, *ACS Energy Lett.* **2024**, 9, 2222.
- [13] a) Y. Gao, S. Zhang, F. Zhao, J. Wang, J. Zhou, W. Li, S. Deng, J. Fu, X. Hao, R. Li, X. Sun, *ACS Energy Lett.* **2024**, 9, 1735; b) L. Hu, J. Wang, K. Wang, Z. Gu, Z. Xi, H. Li, F. Chen, Y. Wang, Z. Li, C. Ma, *Nat. Commun.* **2023**, 14, 3807.
- [14] X. Lin, Y. Zhao, C. Wang, J. Luo, J. Fu, B. Xiao, Y. Gao, W. Li, S. Zhang, J. Xu, F. Yang, X. Hao, H. Duan, Y. Sun, J. Guo, Y. Huang, X. Sun, *Angew. Chem., Int. Ed.* **2024**, 63, 202314181.
- [15] H. Kwak, J. Lyoo, J. Park, Y. Han, R. Asakura, A. Remhof, C. Battaglia, H. Kim, S.-T. Hong, Y. S. Jung, *Energy Storage Mater.* **2021**, 37, 47.
- [16] G. Jia, Z. Deng, D. Ni, Z. Ji, D. Chen, X. Zhang, T. Wang, S. Li, Y. Zhao, *Front. Chem.* **2022**, 10, 952875.
- [17] X. Lin, S. Zhang, M. Yang, B. Xiao, Y. Zhao, J. Luo, J. Fu, C. Wang, X. Li, W. Li, F. Yang, H. Duan, J. Liang, B. Fu, H. Abdolvand, J. Guo, G. King, X. Sun, *Nat. Mater.* **2025**, 24, 83.
- [18] a) S. Wang, Q. Bai, A. M. Nolan, Y. Liu, S. Gong, Q. Sun, Y. Mo, *Angew. Chem., Int. Ed.* **2019**, 58, 8039; b) P.-J. Lian, B.-S. Zhao, L.-Q. Zhang, N. Xu, M.-T. Wu, X.-P. Gao, *J. Mater. Chem. A* **2019**, 7, 20540.
- [19] J. Rak, M. Gutowski, P. Dokurno, H. V. Thanh, J. Bl/azewski, *J. Chem. Phys.* **1994**, 100, 5810.
- [20] a) L. Yang, J. Li, D. Zhang, Y. Liu, Q. Hu, *Commun. Chem.* **2024**, 7, 141; b) S. Mañas-Valero, V. García-López, A. Cantarero, M. Galbiati, *Appl. Sci.* **2016**, 6, 264.
- [21] X. Chi, Y. Zhang, F. Hao, S. Kmiec, H. Dong, R. Xu, K. Zhao, Q. Ai, T. Terlier, L. Wang, L. Zhao, L. Guo, J. Lou, H. L. Xin, S. W. Martin, Y. Yao, *Nat. Commun.* **2022**, 13, 2854.
- [22] a) W. Zhao, J. Yi, P. He, H. Zhou, *Electrochem. Energy Rev.* **2019**, 2, 574; b) M. J. Deck, Y.-Y. Hu, *J. Mater. Res.* **2023**, 38, 2631.
- [23] L.-J. Lai, T.-C. Chu, Y.-C. Hsu, Y.-H. Chen, *J. Appl. Phys.* **2005**, 97, 113524.
- [24] H. Ikeno, M. Krause, T. Höche, C. Patzig, Y. Hu, A. Gawronski, I. Tanaka, C. Rüssel, *J. Phys.: Condens. Matter* **2013**, 25, 165505.
- [25] M. H. Verkuijlen, P. J. M. van Bentum, O. Zabara, M. Fichtner, A. P. Kentgens, *J. Phys. Chem. C* **2011**, 115, 13100.
- [26] H. Koller, G. Engelhardt, A. P. Kentgens, J. Sauer, *J. Phys. Chem. C* **1994**, 98, 1544.
- [27] R. Steudel, T. Chivers, *Chem. Soc. Rev.* **2019**, 48, 3279.
- [28] A. George, S. Sen, J. Stebbins, *Solid State Nucl. Magn. Reson.* **1997**, 10, 9.
- [29] S. Zhang, F. Zhao, J. Chen, J. Fu, J. Luo, S. H. Alahakoon, L.-Y. Chang, R. Feng, M. Shakouri, J. Liang, Y. Zhao, X. Li, L. He, Y. Huang, T.-K. Sham, X. Sun, *Nat. Commun.* **2023**, 14, 3780.
- [30] T. Minami, A. Hayashi, M. Tatsumisago, *Solid State Ionics* **2006**, 177, 2715.
- [31] a) A. Banerjee, K. H. Park, J. W. Heo, Y. J. Nam, C. K. Moon, S. M. Oh, S. T. Hong, Y. S. Jung, *Angew. Chem.* **2016**, 128, 9786; b) L. Duchêne, D. H. Kim, Y. B. Song, S. Jun, R. Moury, A. Remhof, H. Hagemann, Y. S. Jung, C. Battaglia, *Energy Storage Mater.* **2020**, 26, 543; c) L. Duchêne, R.-S. Kühnel, E. Stip, E. C. Reyes, A. Remhof, H. Hagemann, C. Battaglia, *Energy Environm. Sci.* **2017**, 10, 2609; d) A. Hayashi, K. Noi, N. Tanibata, M. Nagao, M. Tatsumisago, *J. Power Sources* **2014**, 258, 420.